

1. Administrative & Study Identification

Full Study Title: A Randomized, Placebo-controlled, Participant- and Investigator-blinded Study to Evaluate the Efficacy in Reducing Atrial Fibrillation Burden (AFB) as Well as the Safety, Tolerability, and Pharmacokinetics of Oral PKN605 in Participants With Atrial Fibrillation **Short Title/Acronym:** PKN605 for Atrial Fibrillation (Protocol CPKN605A12201) **Protocol Number:** CPKN605A12201 / PR000118549 **NCT Number:** NCT07217067 **Sponsor Name:** Novartis Pharmaceuticals **Investigational Product:** PKN605 (Oral) **Phase of Development:** Phase 2 **Medical Monitor:** Jonathan Piccini, Principal Investigator

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy of oral PKN605 in reducing the amount of time participants spend in atrial fibrillation (Atrial Fibrillation Burden - AFB). **Secondary Objectives:** To evaluate the safety, tolerability, and pharmacokinetics of PKN605 in participants with atrial fibrillation (AF).

2.2 Background & Rationale To address the need for novel, targeted treatments for atrial fibrillation to maintain a steady heart rhythm and reduce the impact of the condition. Unlike typical treatments (e.g., beta-blockers, calcium channel blockers, or anticoagulants), PKN605 offers a targeted oral approach that could provide more personalized treatment options with potentially better outcomes and fewer side effects.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind (Participant- and Investigator-blinded) **Randomization:** Randomized (Participants assigned to receive low-dose PKN605, high-dose PKN605, or placebo)

3.2 Planned Enrollment & Duration Target \$N\$: 165 participants **Number of Sites:** Multiple global sites (including locations in the US, Canada, China, Germany, Netherlands, Singapore, and the UK) **Estimated Study Start:** October 28, 2025 **Estimated Primary Completion:** September 09, 2027

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male and female participants ≥ 18 to 110 years of age. 2. History of at least 2 episodes of AF, with at least 1 episode within the last 12 months (or during screening) documented by a 12-lead ECG, Holter, or other ECG recording method. 3. Participants must meet at least one other condition: a moderate-to-high stroke risk based on CHA2DS2-VASc score, AF for at least 1% of the time on a heart monitor, mild and stable heart failure symptoms, or an elevated NT-proBNP level.

4.2 Exclusion Criteria 1. Permanent atrial fibrillation. 2. History of an AF ablation procedure within the last 6 months without a recurrence of AF at least 2 or more months after the ablation. 3. Concomitant use of prohibited medications (Vaughan Williams class I or III anti-arrhythmic drugs stopped < 7 days before screening; amiodarone stopped < 6 weeks before screening). 4. Severely overweight or underweight, or presence of certain heart conditions such as requiring a pacemaker.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Oral administration of PKN605 at varying dosage levels (low-dose or high-dose) or a matching placebo. **Route of Administration:** Oral **Duration of Treatment:** 24 weeks

5.2 Concomitant & Prohibited Therapy Permitted: Standard baseline stroke prevention therapies (e.g., anticoagulants) depending on the patient's CHA2DS2-VASc score. **Prohibited:** Vaughan Williams class I or III anti-arrhythmic drugs and amiodarone (require a washout period prior to screening).

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
1	Screening	Up to 90 days Informed Consent, Medical History, ECG recordings, baseline labs, confirm eligibility.
2-7	Treatment Phase	Weeks 1 - 24 Monthly clinic visits (6 visits total), Randomization, IP administration, continuous at-home AF monitoring (wearable patch/device), safety blood tests.
8	End of Study (Follow-Up)	Week 28 (~ 1 month post-treatment) Final safety follow-up visit, Vital signs, ECG, exit assessments.

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Reduction in Atrial Fibrillation Burden (AFB) (the total amount of time participants spend in

atrial fibrillation) during the 24-week treatment period.

Measurement Method: Continuous heart monitoring using various ECG devices (including wearable heart monitor patches and home checking devices).

7.2 Secondary & Exploratory Endpoints 1. Safety and tolerability of oral PKN605. 2. Pharmacokinetic (PK) profile of oral PKN605 (e.g., clearance and movement of medicine through the body).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Routine collection via regular monthly clinic visits, continuous remote heart monitoring patch readings, and blood test analyses. **Serious Adverse Events (SAEs):** Reported by investigators to Novartis within 24 hours of site awareness. **Data Monitoring Committee (DMC):** An independent Data Monitoring Committee will regularly assess safety signals and unblinded data throughout the study.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy (AFB) evaluation, and Safety Set for the evaluation of adverse events. **Sample Size Justification:** A target of $N = 165$ is appropriately powered to detect a statistically significant reduction in AFB between the combined PKN605 dosage groups and placebo. **Handling of Missing Data:** Standard handling frameworks via Multiple Imputation (MI) or related models for primary endpoint evaluation over the 24-week period.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Combined remote monitoring through connected ECG devices and scheduled in-person monthly site visits for clinical oversight. **Audit Trail:** Secure database for eCRF capturing, continuous home telemetry device data integration, and site qualification checks.

11. Study Identifiers & Governance

Primary ID: CPKN605A12201 **Registry Number:** NCT07217067 **Sponsor/Lead Organization:** Novartis Pharmaceuticals **Study Title:** PKN605 for Atrial Fibrillation **Official Regulatory Title:** A Randomized, Placebo-controlled, Participant- and Investigator-blinded Study to Evaluate the Efficacy in Reducing Atrial Fibrillation Burden

(AFB) as Well as the Safety, Tolerability, and Pharmacokinetics of Oral PKN605 in Participants With Atrial Fibrillation

12. Clinical Framework (Arrhythmia Specific)

Primary Condition: Atrial Fibrillation (AF) **Diagnosis Threshold:** At least 2 documented AF episodes (one in the last 12 months) + elevated risk indicator (e.g., CHA2DS2-VASc, NT-proBNP). **Medical Methodology:** Non-anti-arrhythmic novel oral targeted treatment (PKN605) versus placebo. **Optimization Target:** Reduction of Atrial Fibrillation Burden (AFB)

13. Study Procedures & Methodology

Management Pathway: Heart rhythm telemetry + monthly clinical supervision. **Education Protocol (HCP):** Site-level training on protocol conduct, PKN605 handling, and telemetry equipment provisioning. **Education Protocol (Patient):** Self-monitoring instruction for using home ECG devices/patches and medication compliance. **Quality Audit Mechanism:** Continuous digital device transmission auditing and drug compliance monitoring.

14. Observation & Follow-Up Schedule

Total Duration: Approximately 41 weeks (12-week screen, 24-week treatment, 4-week follow-up) **Onsite Visit Cadence:** Monthly during the 24-week treatment phase (6 in-person visits) **Remote Monitoring Frequency:** Continuous monitoring via wearable heart patch/ECG devices at home. **Checklist Review Interval:** Regular monthly intervals paired with onsite evaluations.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	Jonathan Piccini	_____	[DD-MMM-YYYY]	Clinical Operations Lead	[Novartis ClinOps Lead]	_____	[DD-MMM-YYYY]
Lead Statistician	[Novartis Statistician]	_____	[DD-MMM-YYYY]				